

BIOL3402 Cell Biology and Cell Technology

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Introduction

This course begins with a discussion of cell biology, covering the Cell, Membrane, Nuclear Transport, Techniques in Plant Cell Culture, Transport across the Plasma Membrane, Ion Channels and Osmotogenetics, Junction, Cytoskeleton, Cellular Adhesion molecules, Cell Migration, Cell Signaling and Techniques in Animal Cell Culture. This course aims to introduce the cellular knowledge. Since this is also a Lab-Lecture based course, this course also provide different culturing knowledge and laboratory skills. Upon completion, students will be familiar with the cell biology and cell technology conceptually and practically.

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1 Lecture 16 & 17 Techniques in Cell Culture

1. Reason of performing cell culture (In Vivo)

(a) To study cell physiology

- Want to get high resolution in single cell
- Can control variable and media
- Make protein

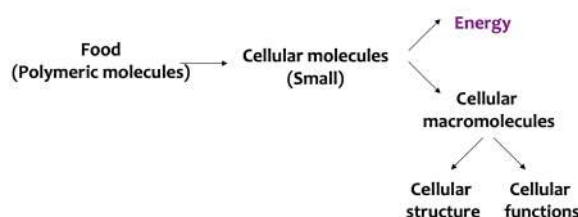
(b) To synthesis valuable cell product

- Gene Cloning + Cell Culture → Recombinant Protein (In Vivo)
- Growth Factor → Change the cell morphology → Differentiate
⇒ **Hepatocyte Growth Factor (HGF)** : Able to see the change of cell morphology
⇒ Cell is **Elongated**
- Major Categories of valuable products
 - Vaccines
 - Monoclonal Antibodies
 - Recombinant Protein : Interferons, Plaminogen Activator, Blood Clotting Factor (Factor VIII)
- Do not use Engineered Bacteria (E. Coli.)
 - ∴ Bacteria do not have appropriate metabolism
 - ∴ Cannot complete post-translational modification in eukaryotic cells

(c) Test the toxicity of a chemical

⇒ Adding directly into culture → Investigate the cell respond

2. Metabolism



- Generate Energy and Macromolecules (Enzymes)

3. Energy

- (a) Enables cellular physiological activities in organisms
⇒ ATPs are needed for muscle contraction
- (b) Essential for biosynthesis
⇒ Synthesis of RNA and DNA driven by ATP hydrolysis
- (c) Activation of biomolecules
⇒ Activation of tyrosine kinase at a specific signaling pathway
- (d) Intermediate compounds in energy pathways are cellular molecules
⇒ **Pyruvate can convert to leucine and valine**

4. Culture Medium Step 1

- (a) Eagle's minimal Essential Medium (EMEM, MEM)
- (b) Reason of using Culture Medium and its Function
 - Used for cell lines
 - Maintain the Consistency
- (c) Composition
 - Glucose
⇒ Carbon Source
 - 13 Essential Amino Acids
⇒ E.g. Glutamine (Unstable) : Act as a precursor for TCA cycle intermediates
⇒ Easy to degrade into toxic byproducts (Ammonia)
⇒ Solution : Use Gibco or Glutamax
 - Vitamins (Low Concentration)
⇒ Utilized as metabolic co-factor
 - Salts
⇒ Ensure the solution is isotonic + No Imbalances with intracellular content and salt (Bicarbonate)
⇒ Maintain the pH range : 6.9 - 7.4

- Antibiotics (Optional)
 - ⇒ Bacteria-specific (Prevent Bacterial Growth)
 - ∴ Reduce the risk of bacterial contamination
- Serum (Optional) : FBS = Fetal Bovine Serum
 - ⇒ Serum is heat-inactivated at 56°C for 30 minutes → Inactivate the complement that may **react with globulin**
 - ⇒ Full of growth factor + Poor Reproducibility (Problem)
- Stem Cells Cell Line : Very Sensitive

(d) Maintaining Physiological pH : **7.4**

i. Examples

- Sodium Bicarbonate (Most Common) : pH7.4 + 5% CO₂
- HEPES 4-(2-hydroxyethyl)-1-piperazineethanesulfonic Acid : pH 7.2 - 7.6
- Phenol Red : pH indicator

ii. Mechanism of Phenol Red

- A. Respiration → More CO₂ → More Carbonic Acid → More Acidic
- B. Acidic : Yellow (Around pH 6.5)
 - ⇒ Too much cell growth
- C. Basic : Purple (Around pH 7.8)
 - ⇒ Cells cannot grow

5. Culture Medium Step 2

(a) MEM → Ham's F12

⇒ Ham's F12 : Contain all component of MEM

(b) Reason of this step : Satisfy some cell lines' requirement

(c) Composition

- All Component in MEM
- Pyruvate (From Glycolysis)
 - ⇒ Additional Carbon Source
 - ⇒ Ensure the energy can be generated

- 7 Non-Essential Amino Acids
- Trace Elements
⇒ E.g. Fe^{2+} , Cu^{2+}
- Organic Compound
⇒ E.g. Adenine, Thymidine, Hypoxanthine
⇒ Precursor for nucleus acid synthesis
- Antibiotics (Optional)
- Serum (Optional)

6. Cons of using **Serum-Supplemented Media**

- (a) Chemically Undefined = High Variation = Inconsistent Result
- (b) Serum is expensive
- (c) Easy to be contaminated by virus

7. Solution from **Serum-Supplemented Media** to Serum-Free Media

- (a) Develop other growth factor (Serum-Free) to replace serum
⇒ E.g. Hormones and Growth Factors
⇒ Specific to special cell lines
- (b) But not all cell lines are able to grow in serum-free condition
- (c) Why still use Serum ?
∴ Others are more expensive

8. Reason for using growth factors-supplemented serum-free media

- Study Endocrine Regulation
- Simply Produce Purification
- Consistency of Reproduction
- Animal Free

9. Growth Factors

- (a) Functions

- Polypeptides stimulate the growth of particular cell types
- Growth-promoting quality rather than its nutritional value
- Not used in metabolic pathway
 - ⇒ But activate signaling pathways
- Effective at **very low concentration**
- Used when we cannot use serum media
 - ⇒ E.g. Endocrine Regulation
 - ∴ Serum contains all kinds of hormones

(b) Common Supplements

- Insulin : Glucose uptake and Metabolism Amino Acid Uptake
- Transferrin : **Iron-Carrying Protein**
- Epidermal Growth Factor : Mitogen for **Epithelial and Fibroblastic Cells**
 - ⇒ Importance : Transferrin > Epidermal Growth Factor
- Other Growth Factors

10. Attachment Factor & Extracellular Matrix

- Some cells need to be sandwiched between cell surface and the substratum (Surface of plastic culture dish) in serum-free condition
- Simulating **in vivo conditions**
- Function 1 : Adhere with other cells
- Function 2 : Make the end sticky → Don't need extracellular matrix
- Example of attachment factors & extracellular matrix
 - Collagen
 - Fibronectin
 - Laminin
 - Poly-lysine

11. Cell Types can be isolated from Animal

(a) Epithelial Cells

- Form monolayer of cell over organ
- Polygonal and Rectangular Shape (3D Shape)
- Easy to culture
- Have relatively rapid growth rate
- Small ratio of Small Nucleus to Cytoplasm

(b) Fibroblasts

- Cells are spindle-like shape on attachment to a solid surface
- Matrix making cells
- Elongated in connective tissue

(c) Muscle Cells

- Myoblast = Precursor cell → Capable of differentiation to form myotubes
- **Myoblasts** can differentiate to form myotubes
- Myotubes can differentiate to form muscle fibers

(d) Nerve Cells

- Highly differentiated + Not divide in Culture
- Precursor Cells $\xrightarrow{\text{Differentiated}}$ Nerve Cells $\xrightarrow{\text{Experiment}}$ Throw the cell (Cannot Reuse)

(e) Blood Cells

- Culture in suspension
- E.g. Lymphocytes
- Very difficult to take image
∴ They float around
- No nucleus → Not capable of replicating

(f) Platelets

- Need to put into reactor + Push them around like in blood
- If can culture → Can do clotting transfusion

12. Primary Cell Culture $\xrightarrow{\text{Very Few}}$ Continuous Cell Line

- Cells are directly derived from animal tissue

- Primary Cell Culture : Freshly isolated cells in culture
- Most cells die within days or weeks (Depending on cell types)
⇒ Die is a defense against bad stuff (Cancer)
- A very small number of cells **continue to grow through multiple passages**
⇒ Finite number of division : Prevent Overgrowth
- Immortalized : Unlimited Proliferation of continuous cell line
⇒ E.g. Tumors, Rodents Cells, Hela Cells (Not cable to Human)
⇒ Requires Transformation : Cells would lose sensitivity to stimuli related to growth control

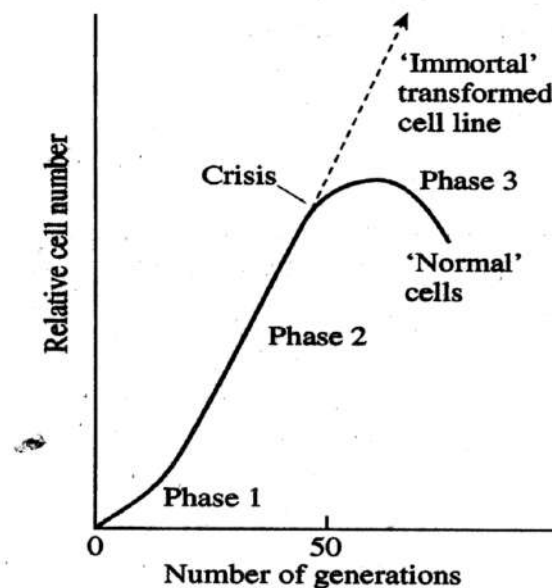


Figure 1: Cell Generations

13. Transformation

- Transformed Cells $\neq$ Cancer Cells
⇒ Malignant : Have ability to form tumor in animals
- We want this in experiments
∴ It is immortalized

- ⇒ But still throw after a long passage number
- ∴ Maintain Optimal Culturing Environment
- May lose their anchorage dependence (Context Dependent)
 - ⇒ Originally : Attach on the substratum
 - ⇒ Now : Able to grow in suspension
- Lose contact inhibition feature
 - ⇒ Can form multi-layers of cell instead of a monolayer



Figure 2: Contact Inhibition

- Chromosome Fragmentation : Some cells might become non-diploid cell with an increase in chromosome number
- High capacity for growth : Even in simple culture media and Without growth factor supplements

14. Method of cells be Transformed

- Spontaneous
- Mutagens : Cause random mutation
- Virus : Infection by retrovirus express activated oncogenes
 - ⇒ E.g. Ras in ERK Pathway

15. Growth Parameter & Steps

- (a) Cell line obtain from the culture collection
- (b) Inoculate cells in growth medium at a density of $10^4 - 10^5$ cell / cm²
- (c) Cell stop growing by some reason (After Several Days)
 - Limited Nutrient
 - Toxic Metabolism Accumulated
 - Lacking Growth Surface

- (d) Sub-culturing / Passaging
 - Trypsinize Monolayer
 - Dilute Cells : 1:2 - 1:10 in fresh media
 - (e) Count Passage Number : Indicates the number of sub-cultures performed
 - (f) New cultures can be established by inoculating some cells into fresh culture medium
16. This growth process requires different density, depending on the cell type
⇒ Moderate Density : Grow very slow, almost no growth
17. Four Phase of a culture
- (a) Lag Phase
 - No Growth in cell concentration
 - Duration : Depends on medium formulation + Initial Concentration
 - Function : Synthesis of growth factors
 - (b) Growth / Log Phase
 - Exponential Increase in cell number
 - Used to determine the doubling time of cell line
$$N = N_0 \cdot 2^x$$
 - N = Final Cell Concentration
 - N_0 = Initial Cell Concentration
 - x = Generation Numbers of Cell Growth
 - Doubling Time = $\frac{\text{Total Time}}{x}$
⇒ Depending on the generation number of cell growth
 - (c) Stationary / Plateau Phase
 - No increase or decrease in cell concentration
⇒ Cell Growth = Cell Death
 - Cell growth is limited

- No more nutrient → Unable to support further cell growth
- Accumulation of toxic metabolic byproducts
- No free growth surface
- Cell stop growing : Single monolayer covers the available substratum (Confluence) → Completely covered in cells = 100%
- But!!! Cell are still metabolically active

(d) Decline phase

- Cell Death > Cell Growth
- Viable cell concentration decrease
- Two Pathways
 - Apoptosis : Intensionally Die
 - Necrosis : Lack of nutrient / Toxic Byproduct
- Change medium can avoid the cell death
∴ Provide food and space

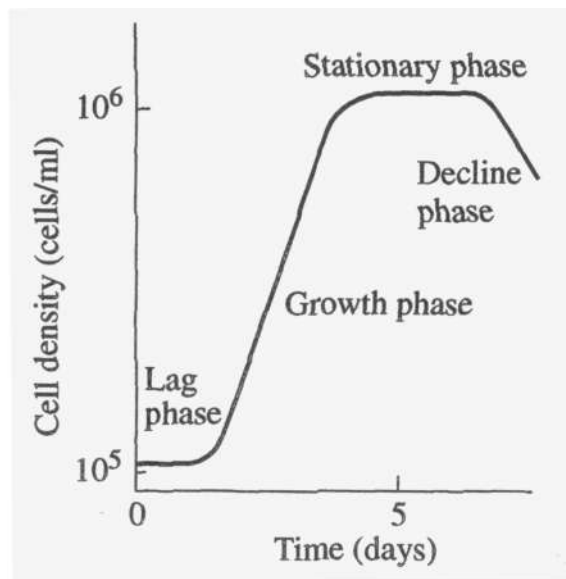


Figure 3: Cell Density

2 Lecture 18 & 19 Techniques in Animal Cells

1. Modification of cells in Culture

(a) Cell Fusion

- (b) Expression of cloned gene in cultured cells
⇒ Transfection / Viral Transduction

2. Cell Fusion

- Somatic cell hybrids are produced by forming heterokaryotic bodies through cell fusion
⇒ Combine their characteristic
⇒ Fused the Nucleus
- Function 1 : Antibody Production
- Function 2 : Chromosomal Localization
- Two Type of Technique : Fusogen (E.g. PEG) vs Electrofusion

3. Fusogen

- Work in diploid cells
- Function : Destabilizes cell membrane → Fuse cell together
- Function : Fuse together by applying voltage
- Pathway : Two Cells (Diploid) $\xrightarrow{\text{Membrane Fusion}}$ Heterokaryon
 $\xrightarrow{\text{Nuclear Fusion}}$ Hybrid Cell (One Fused Nucleus) $\xrightarrow{\text{Mitosis}}$ Cell Line
- Nucleus Fusion : Elimination of Chromosome

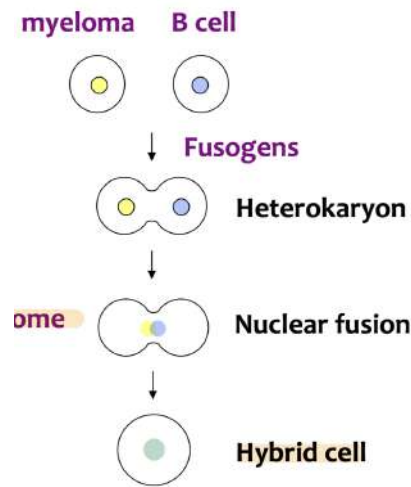


Figure 4: Fusogen

4. Electrofusion

- Two Types : Non-specific vs Specific
- Non-specific
 - Applying current in low voltage : Generate Electric Field
 ⇒ Electric Current is applied in low voltage pulse for short time period
 ⇒ Cell can line up randomly along the line of the current
 - Then line up and aligned by high voltage with long duration
 - Myeloma and B cell will **fused** together in high voltage
 ⇒ Become Viable Hybrid Cells
 - Then Proliferate and replicate itself
 - Do this in sterile chamber
- Specific (Biotin-Avidin System)
 - Biotin attached to **myeloma cells**
 - Avidin linked to **antigen**
 - Then, Antigen binds to specific B cells
 - Biotin binds to avidin with **high affinity**
 - Then, two cells electrically fused together
 ⇒ Form Hybrid Cells

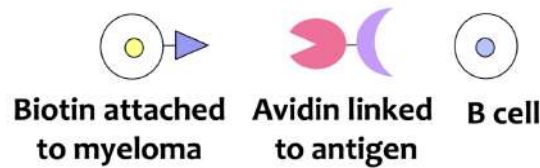


Figure 5: Specific Electrofusion

5. Application of Cell Fusion Technique

(a) Antibody Production

- Normal Antibody-producing Lymphocytes : Limited Life Span
⇒ Cannot grow up infinitely
- Myeloma : Infinite Growth Capacity
- Combine Lymphocytes & Myeloma $\xrightarrow{\text{Fusion}}$ A Mixed Population of Cells (Lymphocytes + Myelomas + Hybridomas)
- Get good hybridomas undergo **HAT Selection**
- Then, Unlimited antibody production is formed
- Purpose : Combine the best antibody synthesis of the B-lymphocyte with infinite growth of myeloma
⇒ Can make more better antibodies

(b) Chromosome Mapping

- Fuse Human & Mouse cells
- Human Chromosome are lost from the hybrid cell lines
- DNA Marker : Perform PCR / Hybridization
- Condition : Myeloma should be able to produce antibodies

6. Expression of cloned gene in cultured cells (DNA Editing)

(a) Transfection (Transfer Naked DNA)

- Calcium Phosphate Method
 - Calcium Phosphate = Positive Charged
 - DNA = Negative Charged

- DNA co-precipitate with $\text{Ca}_3(\text{PO}_4)_2$
 - Neutralize before get into the cell
- DEAM-dextran
 - DEAE-dextran : Positive Charged
 - DNA = Negative Charged
 - DNA complexed with the DEAE-dextran
 - Neutralize before get into the cell
- Liposome
 - Positive Charged Lipid
 - DNA = Negative Charged
 - DNA are packed in liposome vesicle
- Microinjection
 - Insert DNA by a microneedle
 - Insert into the nucleus of individual cell
- Electroporation
 - Promote the uptake of DNA by electrical impulse
 - Applied a voltage
 - ⇒ Disturb the membrane
 - ⇒ DNA can enter

(b) DNA Transfer by Viral Transduction

- Require a plasmid before integrate DNA into cells
- Reason 1 : Highly Replicated the required gene
- Reason 2 : Virus integrate efficiently into the animal cell genome
- Example of viruses for infection : Adenovirus, Retro-viruses, Leutiviruses

7. Adenoviruses & Adeno-Associated viruses

- Both do not get integrated in the genome
- Adeno-Associated Viruses : not pathogenic to human
 - ⇒ Cannot replicate by themselves
- Require specific receptors
 - ⇒ Bind to ligand → Allowing virus enter the cell

- Works in non-dividing cells
8. Selectable Markers : A gene that for hybridomas selection and stable transfection
9. Biosynthesis of Nucleotides
- (a) Two pathways
- De Novo Biosynthesis
 - Salvage Pathway
- (b) De Novo Biosynthesis
- Folic Acid : Byproduct of metabolism
⇒ Synthesis of purine and pyrimidine
 - Folic Acid → Dihydrofolate → Tetrahydrofolate
 - Purine Synthesis : Adenine & Guanine incorporate glycine and tetrahydrofolate
 - Pyrimidine Synthesis : Tetrahydrofolate & Aspartate are raw material
 - **Aminopterin** : Block the conversion of dihydrofolate to tetrahydrofolate
⇒ Inhibits the synthesis of DNA
 - If Aminopterin occurs (Block de novo biosynthesis)
⇒ Cells use another pathway for the synthesis of nucleotide
⇒ Salvage Pathway : Hypoxanthine & Thymidine is needed

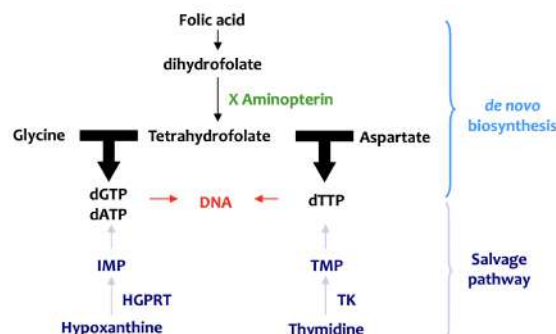


Figure 6: Biosynthesis of Nucleotides

10. Selection System for Hybridoma

- Pathway is mentioned at point 5
- TK Marker Gene : Bromodeoxyuridine (Specific for the thymidine)
⇒ Positive = Die ; Negative = Survive
- HGPRT Marker Gene : 8-Azaguanine (Specific for the recycle purines)
⇒ Positive = Die ; Negative = Survive
- Hybridoma : Positive TK and HGPRT = Survive

11. Selective System : Neomycin Resistance Gene (Neo^R)

- Neo^R Gene : For Aminoglycoside Phosphotransferase
⇒ Able to replicate in medium containing G418
⇒ Form Cell Clones
- Resistance to aminoglycoside group of antibiotics such as G418
- G418 : Prevent replication of mammalian cells

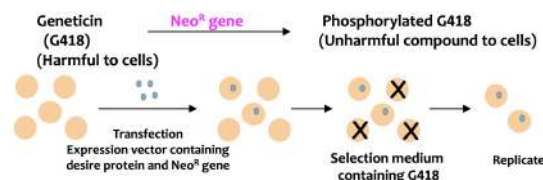


Figure 7: Neo^R Gene

- Puromycin is more efficient for mammalian selection than G418
⇒ But not good for bacterial selection

12. Stable Integration of DNA

(a) Spontaneous (Random)

- Pros : Can be done with almost any vector
⇒ Standard plasmid is enough
- Low Copy

- Cons : Unreliable / Inefficient

(b) Viral

- Pros : Efficient Transduction
- High Copy without random integration
- Cons 1 : Limited Size
 - ⇒ Too much → Efficiency goes down
- Cons 2 : Gene Repression
- Cons 3 : Highly Variable Copy
- Example
 - Retrovirus
 - Leutivirus

(c) Transposon

- Pros 1 : Defined Copy Number
- Pros 2 : High Limited Size
- Limited Copy Number

13. Stable Transfection : Spontaneous

- Clone the best gen into the vector containing Neo^R
- Generate a "Kill Curve" for some cell line
 - ⇒ Find out the amount of G418 for selection
 - ⇒ Different cell cause different amount of antibiotic to kill
- PCR + Western Blot : Check the gene is overexpressed in clones
- Subculture the clones that survive in medium that containing G418

14. Stable Transfection : Viral Integration

- Retrovirus
 - Diving cells
 - Can only infect cells when it is dividing
 - Packaging limit : 7KB
 - Integrate near transcription sites

- Lentivirus
 - Packaging Limit : 10 KB
 - Random Integration
- Use a host cell to generate virus
 - Filter Out Host Cells
 - Polybrene to enhance entry

15. Stable Transfection : Transposons

- Jump from one DNA to another DNA
- Insert in intergenic region
- Long Terminal Repeats (LTR) mark sequence for transposing
- Have different target sequence
- Used for screens
- Less immunogenic than viral method
- Insertion become permanent when transposon expression is lost

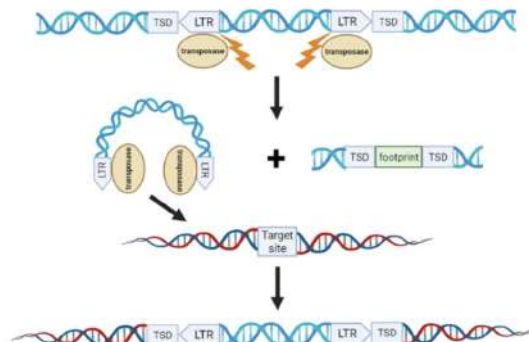


Figure 8: Transposons

16. Laboratory Facilities

(a) Water Source

- Ultrapure Water : Cannot store + Immediate Use
- Can recycle : Minimize algal growth

- Steps
 - i. Soft water is entered into First Stage
 - ii. Hard water is entered into First Stage by Softener
 - iii. First Stage : Distillation / Reverse Osmosis
 - ⇒ Remove Faults
 - iv. Second Stage : Carbon Filtration
 - v. Third Stage : High Grade Deionization
 - vi. Forth Stage : Micropore Filtration
 - ⇒ Forth Stage can recycle back to Second Stage
 - vii. Ultrapure Water is ready for media
- Stage 1 : RO System
- Stage 2-4 : Milli-Q H₂O System

(b) Glassware / Plastic Ware

- Autoclave + Washing is needed
- Cons : Bad refractive index in measurement comparing to plastic
- Pre-sterilized can used in cell culture
 - ⇒ Reduce Contamination

(c) Laminar Flow Hood

- Horizontal Flow Hood
 - i. Air filtered through HEPA filter
 - ii. Air flows out from the hood
 - iii. Create an isolated area
 - iv. Only can use when the culture is safe
 - v. Cannot use to handle primate cell lines
 - vi. Cannot protect the operator

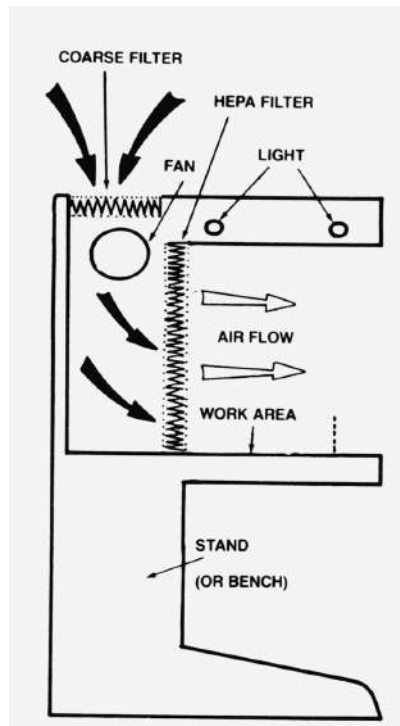


Figure 9: Horizontal Flow Hood

- Vertical Flow Hood
 - i. Vertical flow of air by HEPA filter
 - ii. 70-80% of the air is re-circulated to form air curtain
⇒ Maintain sterile operation and protect the operator
 - iii. Example 1 : Class II Cabinet
⇒ Can work with moderate toxic and infectious agent
 - iv. Example 2 : Class III Cabinet
⇒ Handle high risk pathogens

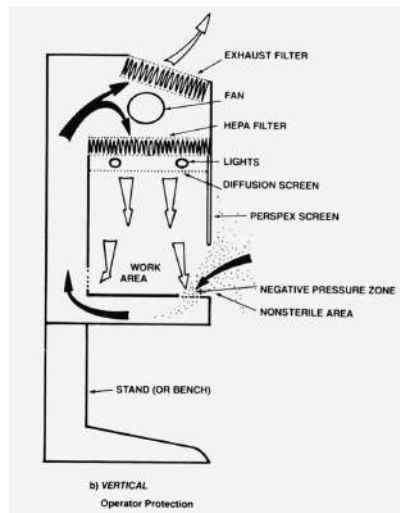
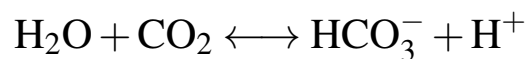


Figure 10: Vertical Flow Hood

(d) Water Jacketed CO₂ Incubator

- Used to maintain the optimal temperature and pH for cell growth
- Water Jacket / Air Jacket : Provide a Constant Temperature Environment
- Contain Bicarbonate-CO₂ Buffering System



- Level of CO₂ gas phase of the incubator : 5%
- pH Level : 7.4

(e) Microscope

- Standard Binocular Microscope : Require the specimen placed on a glass slide + Covered by a cover slip
 ⇒ Need to kill the sample
 ∴ Need to remove a small sample from the culture
- Cons of traditional light microscope
 - i. Temperature & Oxygen Content of sample may change quickly
 - ii. Organism will be under high pressure and unnaturally confined space (Cover Slip)

- iii. Quickly dry out unless replenished with water
⇒ Loss by evaporation
- iv. Replenished water might change the salinity of the sample frequently
- v. Salinity change create stress on microorganisms
- Inverted Microscope : Have the light source on the top
⇒ Enable large object to be placed on the stage

(f) Sterilization and Disinfection

- Kill and Remove microorganism
- Method of Sterilizing the Sample
 - i. Autoclaving
 - Steam Heat : 121°C at 1 kg/cm² for 20-30 minutes
 - Dry Heat : 180°C for 20 minutes
 - ii. Filtration
 - Cellular Ester Filters
 - Removing bacteria + Fungi
 - iii. Irradiation
 - **Ultraviolet Light (UV)**
 - * Wavelength : 260 nm
 - * Acts on nucleic acids
⇒ Strand Breakage, Strand Cross-Linkage, Formation of Pyrimidine Dimers
 - * Inactivate micro-organisms and virus
 - * Act on nucleic acid
 - * Poor Penetrating Power
 - **Gamma Rays**
 - * Induce **covalent bond** under γ -ray
 - * Good Penetrating Power
 - * Acts on nucleic acids
 - * Produce free radicals that kill bacteria
 - * Used when sample cannot be heat-sterilized
⇒ E.g. Plastic Wares

iv. Chemicals

– Gases

- * Ethylene Oxide & Formaldehyde Gas
- * Can kill all microorganism and virus
- * acts on Nucleic Acid & Protein Components
- * Formaldehyde Gas : Decontaminate & Sterilize Laminar Flow Cabinet
- * Both are very toxic

– Solution (Disinfection)

- * Alcohol : 70%
- * Cannot kill fungal spores
- * Bleach : Corrosive

(g) Culture Contamination

- From Culturing Techniques
- Source : Medium, Serum, Tissue, Organ, Equipment
- Example : Bacteria, Mycoplasma, Fungi, Virus

17. Cryopreservation + Freezing (-196°C) + Thawing Cells (37°C)

(a) Storage at low temperature

- -196°C = Liquid Nitrogen
⇒ Most of the time
- -120 - 130°C = Liquid Nitrogen Vapor
- -70°C = Dry-ice + Deep Freezer (Unstable)
⇒ Only short period
- Moderate Cooling vs Rapid Warming in 1 - $2^{\circ}\text{C}/\text{min}$

(b) Physical changes during freezing

- Slow : Dehydration
- Fast : Formation of Intracellular Ice Crystal
⇒ Might damage the cell

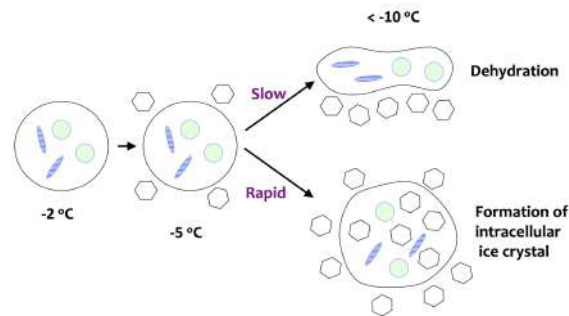


Figure 11: Intracellular Crystalization

(c) Cryoprotectants

- Small molecules are permeable to cells
- Glycerol + DMSO (Toxic to some cells)
- Reduce Intracellular Freezing
 - $\Rightarrow$ Some species cannot freeze
 - $\Rightarrow$ E.g. Fruit Flies
- Increase Extracellular Fluid Volume during extracellular ice formation

(d) Cryo-injuries due to low thawing rate

- Slow Warming : Recrystallization of Intracellular Ice
- Some are toxic to cell at warm temperature

(e) Procedure to freeze and thaw cells

- Freezing
 - i. Resuspend Cells in freezing medium
 - ii. Aliquot into freezing vials
 - iii. Freezing container containing isopropanol
 - iv. -70°C Freezer overnight in the speed $-1^{\circ}\text{C}/\text{min}$
 - v. Liquid Nitrogen Tank
- Thawing
 - Path 1
 - i. Prewarm Medium to 37°C
 - ii. Prewarmed Medium
 - Path 2

- i. Pick one frozen vial from liquid N₂ Tank
 - ii. Thaw the vial at 37°C water Bath
- Centrifugation
 - Discard medium containing DMSO
 - Resuspended cell in fresh medium
 - Seed cells in culture dish

3 Lecture 20 & 21 Techniques in Cell Biology

1. Some cell is very small : Smaller than visible light wavelength
⇒ Use electron microscope → Map out the image
2. Limits of light resolution (Light Microscope)
 - Resolution : Resolving power of microscope
⇒ Depends on the width of the cone of illumination

$$\text{Resolution} = \frac{0.61\lambda}{n \sin \theta}$$

- θ = Half the angular width from a point in the central region of specimen
- n = Refractive index of the medium
- λ = The wavelength of light used
- Numerical Aperture : $n \sin \theta$ = Light-collecting ability
⇒ Dry Lenses : Cannot higher than 1
⇒ Oil Immersion Lenses : High as 1.4
∴ Oil Refractive Index > Water Refractive Index > Air Refractive Index
- ↑ Numerical Aperture , ↑ Resolution + ↑ Brightness of Image

LENSES

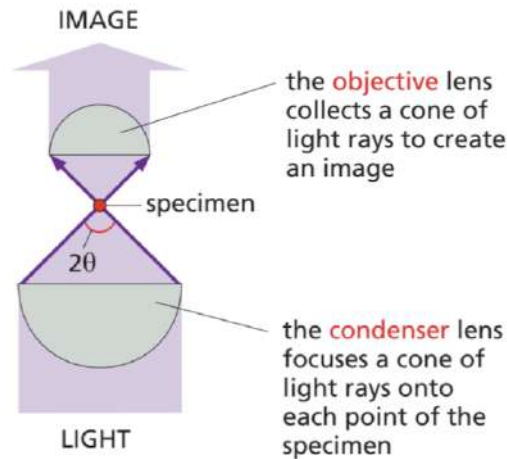


Figure 12: Lenses

3. Objective Medium

- Air
 - (a) Bad when we collect a lot of photon that have high resolution
 - (b) Most common and convenient medium
 - (c) Cheap
 - (d) Good working distance
- Oil
 - (a) Relatively Highest Resolution
 - (b) Relatively best working distance
 - ⇒ Silicone Oil : Low Numerical Aperture, Better working distance
 - (c) Relatively best brightness
- Water
 - (a) Not viscous
 - (b) Might cause contamination if it is not in a stable environment

(c) Do not need to be sterile

4. Transmitted Light Microscopy : Contrast

- Bright Field
- Phase Contrast
 - (a) Use a series of polarizer + prism
⇒ Change the refractive index
 - (b) Bright : Two waves in phase = Constructive Interference
 - (c) Dimmer : Two waves out of phase = Destructive Interference
- Differential Interface Contrast
 - (a) Use prism
 - (b) Adjust the angle to control the contrast (Light Entering)
- Dark Field

5. Fluorescence Microscopy

- Measuring the radiation (Labeling)
- toxic to the cell
- Allows imaging of specific proteins with a high level of contrast
 - ⇒ Genetically encoded fluorescent protein
 - ⇒ E.g. GFP from jellyfish
- Requires 30-120 minutes for the folding of correct 3D shape and develop its ability to fluoresce
- Do not produce light, but it emits light at a different wavelength
- Put Energy in (Absorbed Photon) → Absorbed by fluorescent molecules → Excited State → Emits Photon at longer wavelength → Ground State
- More photon, Shorter wavelength
- The microscope not excited all at the same time
⇒ Avoid too much we can see

6. Mechanism of Fluorescence Microscopy

- (a) First Barrier Filter : Receive (Blue 450-490nm) lights from source
- (b) Beam Splitting Mirror : Reflect (Blue) light with wavelength below (510 nm)
⇒ Transmits light above (510 nm)
- (c) Second Barrier Filter : Cut out unwanted fluorescent signals
⇒ Produce higher wavelength (Green) light emission

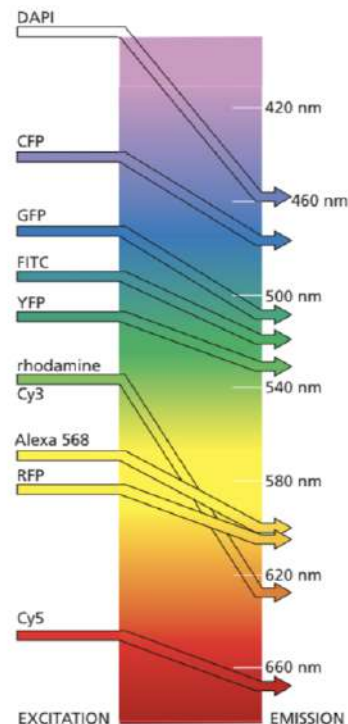


Figure 13: Fluorescence Microscopy Wavelength Emission

- 7. **Plugin Protein** : Replace the toxic radiation
⇒ Can genetically engineer the cell to express and fit it on the protein
- 8. Choice of Fluorophore
 - Fluorophore : Stick to protein by fusion

- Emission / Excitation
- Quantum Yield : How many photon get out for and how many put in
⇒ Always less than 1
- Bleaching
- Maturation Time
- Chromophore
- Oligomerization State

9. Molecular Sensor : Forster Resonance Energy Transfer (FRET)

- Blue Fluorescent Protein emission affects another fluorescent protein's emission
- Yellow Fluorescent Protein emits yellow light when there is Ca^{2+} input
- Vice Versa when cAMP input
⇒ i.e. Blue Fluorescent emits blue light

10. Microscopy in 3D

- Look at non-plane light
- Method : Blocking / Selective Excitation

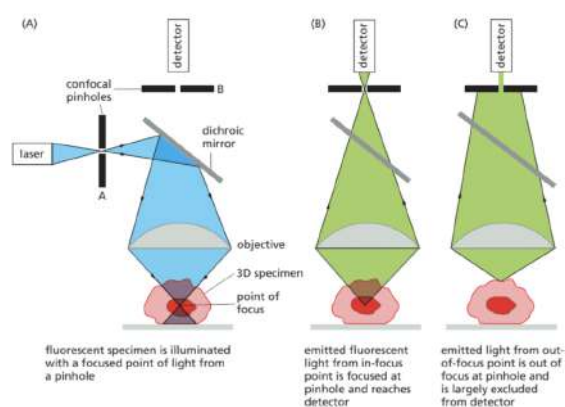


Figure 14: 3D Microscopy

11. Blocking

(a) Confocal

- Point Scanner
- Adjustable Pinhole
- Adjustable Zoom
- Higher Image Quality
- Large Field of View
- Can Use Scanner for Stimulation

(b) Spinning Disk

- Faster
- Small field of view
- Small Artifacts
- Cannot Adjust Pinhole
- Minimize exposure time based on disk speed

12. Selective Excitation

(a) **Multiphoton Excitation**

- Better Penetration In Vivo Imaging
- Lower Phototoxicity
- Improved z-resolution
⇒ Scales with number of photons
- Requires Specific Fluorophores
- Specialized Lasers
- Used for stimulation

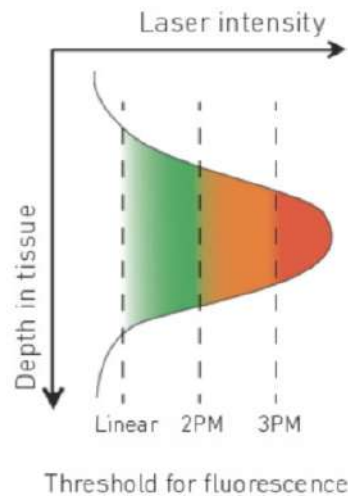


Figure 15: Threshold for Fluorescence

(b) Total Internal Fluorescence Microscopy (**TIRFM**)

- Can only image within 100 nm of the coverslip
- Limited high zoom and small fields for objective based techniques
- Can image extremely fast with low toxicity
- Use total internal reflection (Almost no bleaching)

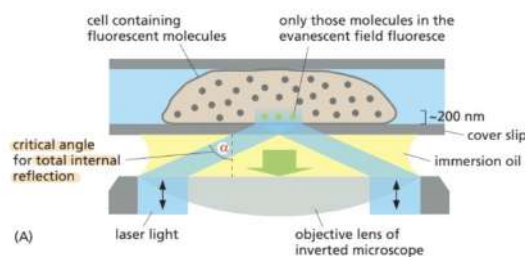


Figure 16: Mechanism of TIRFM

(c) Lightsheet Microscopy

- Fast
- Low bleaching
- Lots of postprocessing
- Require Restrictive Mounting

- Thick Tissues : Uneven illumination

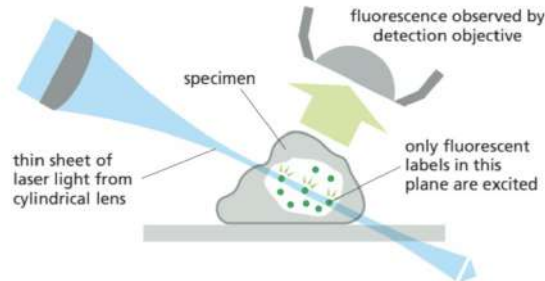


Figure 17: Lightsheet microscopy

13. Superresolution Microscopy

- (a) Measuring things smaller than the light wavelength + Differentiated Limit

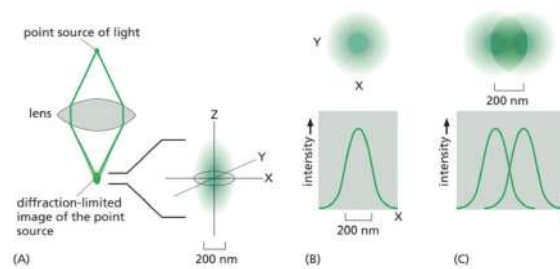


Figure 18: Superresolution Microscopy

14. Cons of Superresolution

- Requires Specific Dyes / Fluorophores
⇒ Need to be very bright and stable
- Most cannot use on living sample ⇒ Usually slow + Need many images to make one image
- Phototoxic
- Easy to be Bleached
- Sample cannot move
- Need to be very stable
⇒ Small variation → Ruin Image

15. Electron Microscopy

- Fixed Sample only
- Not only small object can be used
 - ⇒ Larger Objects : Can get a better image than light microscope
 - ∴ Cell cannot block electron
- Need Metal Coating
 - ⇒ To get High Contrast
 - ⇒ The biological samples are mostly made of light atoms (C, H, O, N) → Don't scatter electrons well → Lower Contrast
- Have Sub-molecular Resolution
 - ⇒ Can resolve the shapes and arrangements
- Sample need to be frozen to preserve state